

Astana IT University

ArtGuard

Risk Assessment Prototype and Investment Pitch for a Blockchain Startup

Introduction to Finance
Final Project
Team 03

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GitHub repository: https://github.com/ONI-COLLI/Artguard

University OneDrive video: upload and institutional sharing pending

Network: local Anvil, chain ID 31337. No public testnet address.

Educational simulation. Fictional clients, planning figures and test assets only. No real investment or financial advice.

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Main text consists of exactly 15 pages from Introduction through Conclusion. References and appendices are outside this count. Body text is Times New Roman 12 pt with 1.5 line spacing and 2.5 cm page margins.

All financial forecasts are educational assumptions. The website's live display currencies, custodial test-wallet balances and demo transaction totals are separate from the commercial financial model.

Release status: the public GitHub URL is recorded. University OneDrive sharing, individual contributions and student review remain required before submission. No working institutional URL is invented.

1 Introduction

ArtGuard is an educational digital-art licensing marketplace. A creator uploads an original file, registers its SHA-256 fingerprint and offers personal, commercial or extended usage rights. The buyer deposits test ETH into escrow. Acceptance activates a wallet-bound licence and credits the platform, creator and co-authors. The project demonstrates a technical process. It has no real customers, investment commitments or monetary revenue.

The Final Project evaluates two questions separately. First, can the core licence and payment workflow be reproduced and independently checked on a local Ethereum Virtual Machine? Second, could a hypothetical Kazakhstan service cover its costs under explicit commercial assumptions? Working software answers the first question. It does not by itself establish customer demand or an attractive investment return.

Table 1. Changes since the Endterm and the earlier report draft

Earlier assumption or weakness	Final Project response
Immediate settlement and a fixed 90/10 split	Escrow, delivery and review deadlines, co-author shares, and 10/5/0 plan commissions.
Hardhat, PostgreSQL and a different contract described as implemented	Document the actual Anvil, SQLite and v4 escrow implementation.
Coverage percentages and gas averages without matching evidence	Remove unsupported measurements and attach actual test and receipt evidence.
A short payback based on incomplete costs	Include salaries, security, processing of full buyer payments and downside funding needs.

Source: Team comparison of the supplied Endterm summary, earlier Final draft and current repository.

Table 1 summarises the revisions. The report retains the team's draft customer targets and spending assumptions for comparison. It changes the commission calculation to match the product and corrects the processing-fee base. These revisions produce a negative three-year NPV. The investment recommendation therefore becomes conditional on validating demand and reducing the cost of serving each transaction.

The objectives are to document and test the prototype, evaluate market, technical, operational, financial and regulatory risks, produce reproducible three-year statements, and assess a hypothetical investment proposal. Sections 2 and 3 separate implemented controls from future safeguards. Section 4 states all planning inputs and shows their consequences. No company registration, asset purchase or real-money activity forms part of the coursework.

2 Prototype and security

2.1 Platform choice and consensus

The implemented prototype uses Anvil, an Ethereum-compatible local development node with chain ID 31337 (Foundry, n.d.). The operator creates blocks and controls the node. Anvil does not demonstrate a production validator network, external decentralisation or public-chain finality. Local execution is sufficient to test Solidity rules and inspect real local receipts.

Table 2 compares Polygon PoS with permissioned BFT. Polygon's Bor execution layer and Heimdall-v2 consensus layer coordinate block production, validation and Ethereum checkpoints (Polygon, n.d.). Consensus selection changes who must be trusted, not just transaction speed (Lashkari & Musilek, 2021).

Table 2. Public PoS and permissioned BFT comparison

Criterion	Polygon PoS option	Permissioned BFT option
Validation	External stake-based validator system	Known approved organisations
Verification	Independent access to public records	Depends on participant access and governance
Cost	Variable gas plus application hosting	Node operations and governance costs
Privacy	Public metadata requires minimisation	Permissioned access helps limit readership
Decision	Candidate for later evaluation	Useful if known institutions share governance

Source: Own comparison based on Polygon (n.d.) and Lashkari and Musilek (2021).

2.2 Actual roles and trust boundaries

A regular account can both publish and buy. Demo buyer and creator accounts are personas, not different permission classes. Managers handle financial reporting and disputes. Technical administrators handle site operations. The contract checks wallet addresses directly rather than using OpenZeppelin AccessControl roles.

The quote signer and arbiter are trusted immutable addresses in v4. A seller confirms delivery, a buyer accepts or disputes, and the arbiter resolves a disputed order. A recipient withdraws its own credit. No implemented multisig, emergency pause or signer-rotation function is claimed. Those are future controls requiring a new design, tests and audit before any commercial use.

2.3 Architecture and the main use case

Figure 1 shows the implemented architecture. Express validates requests and uses SQLite for accounts, per-tab sessions, project metadata and accounting copies. Original files stay in protected server storage. Solidity contracts store registration claims, escrow state, licence validity and withdrawable balances. The server synchronises only after verifying receipts and expected events.

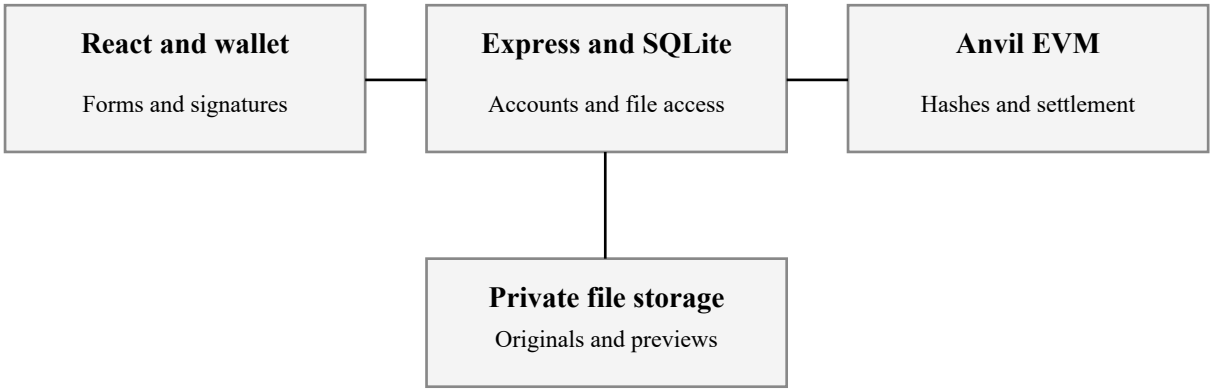


Figure 1. Implemented local ArtGuard architecture

Source: Current React, Express, SQLite and Anvil implementation.

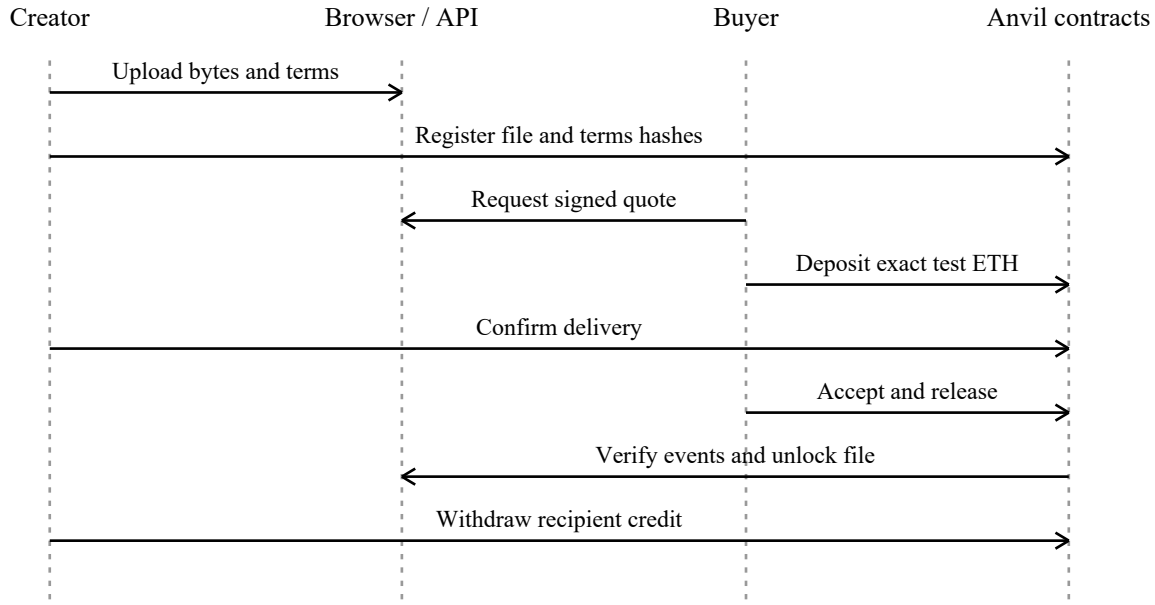


Figure 2. Registration, escrow settlement and withdrawal sequence

Source: contracts/ArtworkRegistry.sol and contracts/EscrowLicenseMarketplace.sol.

Figure 2 starts with file bytes and a normalised terms snapshot. A registration transaction records their fingerprints and the registering wallet. A changed byte normally produces a different file hash. This supports an exact comparison, not visual plagiarism detection or a legal finding of authorship. Digital rights systems still need clear off-chain governance and licensing terms (Ciriello et al., 2023).

2.4 Escrow rules and security evidence

A deposit locks funds and creates neither an active licence nor earned sales revenue. The seller has 72 hours to confirm delivery. The buyer then has 24 hours to accept or dispute. Acceptance releases payment. After review expiry, anyone can submit release for an undisputed order. A dispute blocks ordinary release. The arbiter can decide release or refund, and a refund becomes available after seven unresolved days. Time enables an action; it does not send a transaction.

Table 3. Relevant threats and current controls

Threat	Implemented control	Residual issue and future work
Reentrancy	Credit is cleared before transfer; nonReentrant protects withdrawals.	Independent hostile-recipient and invariant review before launch.
Quote tampering or replay	EIP-712, buyer checks, expiry, instance binding and consumed quote IDs.	The trusted signer can still authorise incorrect terms.
Unauthorised settlement	Seller, buyer and arbiter address checks plus state transitions.	A compromised arbiter can decide a disputed order.
Key loss and operator outage	Disposable local wallets and restricted signer routes.	Multisig, recovery, monitoring and node resilience remain planned.

Source: Own inspection informed by Kushwaha et al. (2022a, 2022b) and OpenZeppelin (n.d.).

Table 3 records actual controls. Ten EVM escrow tests passed in the recorded demonstration and again during this review. They cover invalid quotes, exact splits, withdrawal, delivery and dispute timeouts, manager permissions, restart persistence, legacy licence reads and reconciliation. The video separately shows 21 selected application unit tests. The complete client suite later passed 98 tests. These are distinct runs and scopes. No line-coverage percentage is claimed.

Reproduction uses npm ci, npm run build, npm run test:defense:contracts and npm run demo:defense. The isolated defence page runs on 127.0.0.1:3600. Start real demonstration deploys to its separate Anvil network. Appendix C records the saved run and successful transaction hashes. The planned USD 10,000 audit and subsequent security work are budget assumptions, not a vendor quote or a completed audit.

Public code, contracts and tests: https://github.com/ONI-COLLI/Artguard.

3 Risk assessment

3.1 Likelihood and impact matrix

Risk scores are team planning judgements, not measured frequencies. Likelihood and impact each range from 1 to 5, and the score is their product. A likelihood of 4 means the team considers the event plausible enough to test early. An impact of 5 means it could stop the proposed launch or require substantial new funding. Table 4 covers the required categories. The complete register includes owners, triggers and responses in Appendix D.

Table 4. Main risk matrix

ID and category	Risk	L	I	Score
R1 Market	Too few active buyers and paying creators	4	5	20
R3 Technology	Contract defect or incorrect signed quote	2	5	10
R4 Security	Compromised quote signer or arbiter	3	5	15
R6 Operational	Loss of team capacity or original files	3	4	12
R7 Operational	False authorship claims and delivery disputes	4	3	12
R8 Financial	Funding expires before viable operations	4	5	20
R10 Regulatory	A future payment model needs additional authorisation	3	5	15
R11 Regulatory	Personal-data breach or unsuitable hosting	2	4	8

Source: Own qualitative assessment. L = likelihood and I = impact; score = L × I.

Market and financing risks dominate because technical success cannot create demand. Even the base case has a negative three-year NPV. A missing original file can also break the customer experience despite a valid blockchain licence. The chain cannot reconstruct files that the operator has lost.

Security papers identify recurring vulnerability classes, but using a published checklist cannot establish that this particular deployment is safe (Kushwaha et al., 2022a). The risk register therefore retains key compromise and contract defects after the listed controls. The next funding decision should depend on pilot evidence and independent review, not on the absence of failures in ten tests.

3.2 Kazakhstan regulation and intellectual property

The legal discussion concerns a hypothetical Kazakhstan service. The current coursework uses fictional accounts and test ETH only. ArtGuard has no commercial licence, registered operating company, certified local hosting arrangement or real payment-provider integration. Consent screens and policy pages demonstrate application behaviour; they do not prove legal compliance.

Digital assets require a model-specific assessment under the Law on Digital Assets and, where relevant, the AIFC Digital Asset Activities rules. A licence record, custody of customer funds and exchange services raise different questions. A future operator would obtain advice on the precise activity and jurisdiction before accepting any real asset. An AIFC route cannot be assumed merely because the startup is in Kazakhstan (Law No. 193-VII, 2023; AIFC, n.d.).

A possible first commercial phase would use a licensed provider for tenge payments and keep only appropriate records on a public chain. That fiat path is a proposal, not implemented functionality. It would need new payment integration, chargeback handling, a revised receipt model and tests. Purchasing gas or launching on a main network is outside this educational project.

Personal-data processing would require a defined operator, lawful purposes, an appropriate consent or other applicable basis, access controls, retention rules and review of database location and transfers. The prototype keeps account details off-chain and offers consent history and privacy requests. Wallet metadata can still be linkable to a person. Off-chain storage alone does not make it anonymous (Law No. 94-V, 2013).

Copyright arises through creation, while a hash record supports an evidence trail. It does not grant rights over another person's work. A future operator needs clear licence scope, duration, territorial terms, co-author permissions, takedown and appeal procedures. Expiry can stop new downloads in ArtGuard but cannot erase copies already downloaded (Law No. 6-I, 1996; WIPO, n.d.).

An LLP is the hypothetical legal form retained from the team draft. Founder IP assignments, a trademark strategy and third-party licences would require review before launch. The financial model uses 20% tax and a 17.5% employment-cost burden as simplified planning inputs. They are not statements of the complete statutory tax calculation. The model assumes no preferential tax treatment and does not treat a future registration or authorisation as already obtained.

3.3 Risk responses and their cost

Every paid mitigation must appear in the financial model. Table 5 links security and legal measures to specific months. These figures are explicit budgeting assumptions. Procurement has not occurred and no supplier has promised these prices. Salaries also fund ordinary moderation and incident response, so their cost is not added again as a separate reserve.

Table 5. Mitigations included in the forecast

Measure	Planning cost	Timing and model line
External contract audit	USD 10,000	Month 4; security
Two re-audits	USD 3,000 each	Months 16 and 28; security
Bug-bounty allocation	USD 2,000 annually	Prorated monthly from Month 5; security
Hardware-wallet allowance	USD 300	Month 4; security
Initial legal review	USD 1,500	Month 1; administration
Further legal reviews	USD 1,000 each	Months 13 and 25; administration

Source: Own calculations from scripts/finance-model.mjs. Amounts are hypothetical.

A hardware-wallet budget does not mean a multisig exists. The current v4 signer and arbiter are immutable. A compromise response therefore includes stopping new off-chain quotes, preserving evidence and assessing a replacement deployment. Existing contract rights and credits require careful migration. There is no emergency pause that can simply freeze this contract.

The market response is a staged pilot with consenting, anonymous research participants and fictional demonstration data. Adobe Stock already supplies content-licensing terms, while Pixsy focuses on finding and addressing unauthorised image use. These are competitor categories, not partners. ArtGuard's proposed difference is an integrated licence, escrow and co-author audit trail. Its value to customers still needs testing (Adobe, n.d.; Pixsy, n.d.).

Funding is also a control. The second tranche should require documented pilot retention and unit economics, rather than a transaction count created by demo accounts. If these milestones fail, management should reduce scope, postpone hires or stop the hypothetical project. Those actions change the cost and growth assumptions and require a new model run. They cannot be claimed to solve a funding shortfall automatically.

4 Detailed financial plan

4.1 Forecast assumptions and units

The model covers 36 months, with development in Months 1–4 and sales from Month 5. Table 6 gives demand anchors; Tables A1–A3 show monthly output. Values are hypothetical KZT amounts. A frozen planning rate of 447.94 KZT per USD preserves comparison with the team draft. It is not represented as a verified current exchange rate. Live display-currency conversion on the website is separate and never changes these forecast assumptions.

Table 6. Demand and subscription anchors

Monthly level	Month 5	Month 12	Month 24	Month 36
Licence sales	50	400	1,300	2,500
Premium creators	15	150	350	600
Business creators	2	20	60	120

Source: Demand targets retained from the team draft; no survey or actual customer adoption is claimed.

Expected volumes interpolate linearly between anchors. Fractional monthly values are forecast averages. Average licence price is USD 30. Premium costs USD 10 and Business USD 50 per month. The assumed shares of sales value are 50% Free, 30% Premium and 20% Business. The effective commission is therefore 0.50 × 10% + 0.30 × 5% + 0.20 × 0% = 6.5%. Subscriber counts refer to creators, who can also be buyers.

Processing costs are 2.5% of gross licence payments plus subscriptions, not 2.5% of the platform's commission alone. Gas receives a USD 0.10 allowance per sale. Hosting is USD 100 monthly in Year 1, twice that in Year 2 and three times that in Year 3. Promotion revenue is excluded from the base forecast because uptake has not been validated. The implemented USD 5 seven-day promotion remains a possible upside, not a necessary source of forecast profit.

Four founders receive monthly gross salaries of KZT 300,000, 400,000 and 500,000 across the three years. A KZT 400,000 community role begins in Month 13, followed by a KZT 550,000 developer in Month 25. A 17.5% planning burden applies to gross salaries. Marketing is KZT 100,000 before launch and KZT 300,000/450,000/600,000 monthly thereafter. Administration includes USD 100 tools, KZT 140,000 other monthly costs and the legal reviews in Table 5.

4.2 Income and cash flow statements

Table 7 separates gross marketplace volume from platform revenue; Table 8 reconciles cash. Creator and co-author proceeds are not platform income. The operating model excludes VAT from revenue and does not model VAT remittance, detailed statutory payroll rules or chargebacks. Those are limitations requiring a qualified review before any commercial use. Amounts below are KZT thousand, rounded for presentation.

Table 7. Three-year projected income statement

Item	Year 1	Year 2	Year 3
Gross licence volume	24,189	143,117	314,454
Commission revenue	1,572	9,303	20,440
Subscription revenue	4,927	25,085	50,953
Total platform revenue	6,500	34,387	71,393
Processing fees	728	4,205	9,135
Gas and hosting	618	1,552	2,661
Salaries and burden	16,920	28,200	41,595
Marketing	2,800	5,400	7,200
Security	5,211	2,240	2,240
Administration	2,889	2,665	2,665
EBITDA and EBIT	-22,667	-9,875	5,897
Planning corporate tax	0	0	0
Net income and operating FCF	-22,667	-9,875	5,897

Source: Own calculations from scripts/finance-model.mjs. Amounts are hypothetical.

Table 8. Projected cash flow statement in KZT thousand

Item	Year 1	Year 2	Year 3
Opening cash	0	17,333	7,458
Operating cash flow	-22,667	-9,875	5,897
Investing cash flow	0	0	0
Equity financing	40,000	0	0
Closing cash	17,333	7,458	13,355

Source: Own calculations from scripts/finance-model.mjs. Amounts are hypothetical.

4.3 Break even and investment value

Table 9 places operating break even in Month 23. This is an operating milestone, not recovery of all development losses. Subsequent audit costs or hiring can again make a month negative. Simple payback does not occur within the 36-month forecast, and no unsupported date beyond that horizon is claimed.

The annual discount rate is 25%, within the required venture range of 20–30%, to reflect early demand uncertainty, concentrated execution risk and illiquidity. The model uses year-end discounting of annual free cash flow. Initial development outflows are already in Year 1. Equity financing is excluded from project FCF, so subtracting the KZT 40 million raise again would double count funding.

Table 9. Base-case investment metrics

Metric	Result
Three-year project NPV at 25%	KZT -21.43 million
Maximum cumulative cash need	KZT 35.76 million in Month 28
Simple and discounted payback	Neither reached within 36 months
Cash after financing in Month 36	KZT 13.35 million
Optional terminal value, g = 5%	KZT 30.96 million at end of Year 3
NPV including terminal value	KZT -5.58 million

Source: Own calculations from scripts/finance-model.mjs. Amounts are hypothetical.

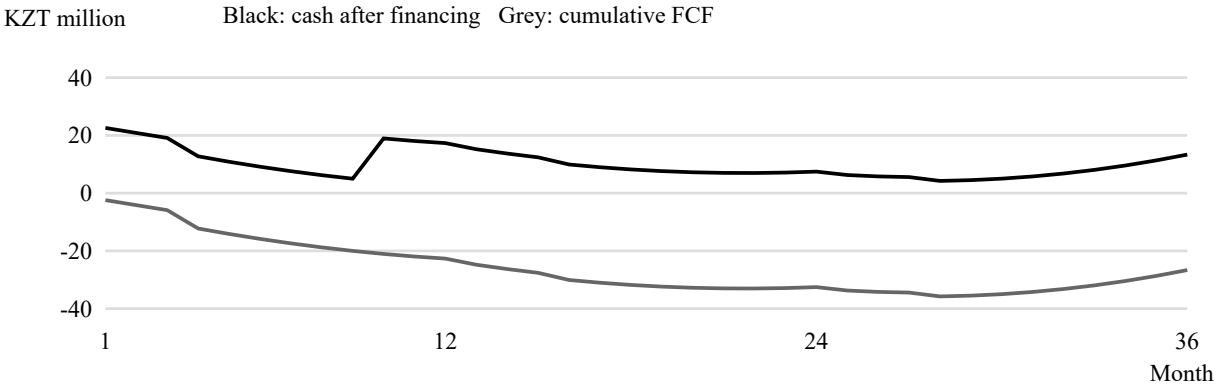


Figure 3. Cumulative project FCF and financed cash balance

Source: Own calculations from scripts/finance-model.mjs. Amounts are hypothetical.

The optional terminal value assumes Year 3 FCF grows by 5% perpetually. It is a sensitivity assumption, not observed business value. Even this extension does not produce a positive base-case NPV. Figure 3 shows positive cash despite the project's negative NPV. The appropriate response is to revisit unit economics and customer evidence, rather than relabel financing as earned income.

4.4 Sensitivity and downside funding

Table 10 compares three downside scenarios. Scenario A reduces all sales and subscriber counts by 30%. Scenario B increases all forecast cash operating costs by 20%, including security and payment processing. Scenario C combines both changes. The model recalculates variable costs, tax-loss balances and monthly cash rather than simply subtracting a percentage from final profit.

Table 10. Sensitivity of NPV, payback and funding need

Metric	Base	A	B	C
NPV Years 1–3, KZT m	-21.43	-37.91	-38.47	-54.41
Cash need, KZT m	35.76	55.70	56.60	82.47
Month of maximum need	28	35	32	36
Extra above KZT 40m	0.00	15.70	16.60	42.47
First positive EBITDA month	23	36	33	Not reached
Payback by Month 36	No	No	No	No

Source: Own model. A: customers −30%; B: cash operating costs +20%; C: both changes.

The base funding requirement is KZT 35.76 million. Lower demand raises the requirement to KZT 55.70 million, while higher costs require KZT 56.60 million. The combined case needs KZT 82.47 million. A KZT 40 million raise cannot finance those downside plans unchanged.

Management responses would include deferring hires, narrowing the product scope and delaying launch. The current stress table does not assume that these responses happen for free or preserve customer growth. Each revised strategy needs an explicit new scenario. A negative projected cash balance means an unfunded plan, not permission to spend money the company does not have.

A subjective risk-weighted estimate can support discussion, but cannot replace evidence. Using the draft weights of 50% base, 25% lower demand, 15% higher costs and 10% combined yields a negative expected three-year NPV. These weights are judgemental and do not make the forecast statistically reliable. The shared result across the scenarios is that customer growth and the recurring cost base must be validated before a full funding commitment.

No terminal value is credited to scenarios with negative Year 3 FCF. This convention avoids applying a perpetual-growth formula to a loss-making year and reporting an implausible asset value. It also shows why a profitable last month alone does not establish a sustainable full-year business.

4.5 Use of funds and runway

The hypothetical raise remains KZT 40 million, with KZT 25 million in Month 1 and KZT 15 million in Month 10. Table 11 shows gross spending to the base-case cash trough in Month 28. The shares refer to spending, not directly to the investment. Customer receipts finance part of those costs. Keeping these bases separate prevents a use-of-funds chart from implying that the same capital is spent twice.

Table 11. Spending to Month 28 in KZT million

Use	Amount	Share of spend
Team salaries and burden	58.98	61.6%
Marketing	10.60	11.1%
Security	9.09	9.5%
Administration	6.74	7.0%
Hosting	2.15	2.2%
Gas and processing	8.19	8.6%
Gross spending	95.76	100%
Less platform revenue	60.00	Not a funding tranche
Cumulative cash need	35.76	Net of receipts
Reserve within KZT 40m	4.24	Remaining buffer

Source: Own calculations from scripts/finance-model.mjs. Amounts are hypothetical.

With no revenue, the full KZT 40 million covers approximately 15 complete months of the model's fixed costs. This runway measure assumes both tranches arrive and removes payment-processing and gas costs because there are no sales. In the base case the minimum financed balance is KZT 4.24 million. The model does not count locked customer escrow deposits as available company funding.

The conditional second tranche creates a further execution risk. The cash forecast assumes it arrives in Month 10. If the milestone fails, the company cannot continue the same spending plan simply because the total proposed raise is KZT 40 million. The finance lead must run a no-second-tranche case before committing to hires or an audit contract.

The earlier draft showed a much shorter payback and lower funding need. The revised model retains its demand anchors and salaries, but applies a 6.5% blended commission and processing fees to full buyer payments. These corrections explain the weaker result. The frozen planning exchange rate is also exposed as a model assumption, while a separate currency-risk scenario remains sensible future work.

5 Investment pitch

5.1 Executive summary

ArtGuard addresses a practical licensing problem. A creator needs to document the work, the permitted use and the agreed price. A buyer needs a clear licence and confidence that payment follows delivery. Co-authors need a traceable account of their agreed shares. A conventional database can support this service, while a public blockchain could make settlement and licence records independently verifiable.

The working prototype combines a catalogue, project uploads, per-project co-author shares, expiring or perpetual licences, subscription tiers, promotion payments and escrow. Its local demonstration records a buyer deposit, seller delivery, acceptance, an active licence and a co-author withdrawal. Ten EVM escrow tests passed. These are technical milestones using fictional identities, not commercial traction.

The proposed first market is independent visual creators and small business buyers in Kazakhstan. No verified market-size estimate or customer survey is claimed. Adobe Stock demonstrates an established licensing category, Pixsy addresses infringement, and NFT research discusses a different token-market model (Adobe, n.d.; Pixsy, n.d.; Bhujel & Rahulamathavan, 2022). ArtGuard must test whether its combined workflow provides enough value to justify an additional service.

Revenue would come from creator subscriptions, plan-dependent sales commissions and optional promotion. Free costs USD 0 with a 10% fee, Premium USD 10 with 5%, and Business USD 50 with 0%. The forecast assumes a 6.5% weighted fee. It excludes unvalidated promotion demand and uses an average licence price of USD 30. Payment processing, support, security and original-file storage remain material costs.

The proposal requests KZT 40 million in two tranches for an illustrative 26% participation interest. The base case reaches KZT 71.39 million revenue in Year 3, but three-year NPV is KZT -21.43 million and payback exceeds the forecast horizon. The implied investment return under an assumed exit is below the 25% hurdle. The current terms therefore require revision or stronger evidence.

The recommended decision is a staged validation project, not an unconditional investment commitment. The first work should test creator retention, buyer willingness to pay, expected dispute cost and unit contribution after payment processing. Only a reviewed model that survives credible downside cases should support the next hypothetical round. No actual investment is solicited by this coursework.

5.2 Funding terms and team

Table 12. Illustrative investment terms and their limits

Item	Proposal or calculation
Funding	KZT 25m initially and KZT 15m in Month 10, subject to evidence-based milestones.
Equity	26% after both tranches; hypothetical terms requiring a formal agreement.
Implied valuation	KZT 153.85m post-money and KZT 113.85m pre-money.
Exit assumption	2.5 × Month 36 annualised revenue, plus cash; an assumption, not a quoted market multiple.
Illustrative outcome	1.56× invested capital and 17.7% annual IRR using the two actual tranche dates.
Decision	The illustrative IRR is below 25%. Reprice, reduce costs or validate a stronger revenue case.

Source: Own calculations from scripts/finance-model.mjs. Amounts are hypothetical.

Table 12 separates investor IRR from project NPV. The exit multiple is uncertain for a small unprofitable company. Downside financing requires a dilution or debt model. A strategic sale or later investor buy-out is an uncommitted exit option.

Table 13. Team responsibilities

Member	Role and review responsibility
Assan Dairabaev	CEO; Introduction, pitch and conclusion
Demezhan Kaidar	CTO; Architecture, contracts, security and technical evidence
Yevgeniy Tyo	CFO; Financial assumptions, model and sensitivity analysis
Kalybek Zhumabekov	COO; Operations, risk and regulatory analysis

Source: Team roles confirmed by the project representative. Each student confirms their own actual contribution separately.

The technical lead must explain quote validation, state transitions and withdrawals. The finance lead must reproduce the model and defend its assumptions. The operations lead owns the risk and regulatory discussion. The team lead connects these findings to the investment decision. Responsibilities do not establish who wrote every line of code: individual GitHub commits and self-evaluations must document the actual work.

6 Conclusion

The prototype demonstrates a complete local blockchain use case. A creator registers exact file and terms fingerprints, a buyer deposits test ETH, and the contract enforces delivery and review conditions. Acceptance activates a licence and credits recipients. A separate withdrawal transfers earnings. Successful receipts, canonical blocks, events and contract state can be inspected without treating an accounting database row as proof of payment.

This technical result has clear limits. The local Anvil operator controls the network. Demo wallets use a custodial test signer. A trusted manager resolves disputes through the arbiter address. File delivery, account controls and original storage remain server responsibilities. A hash identifies bytes and a registration claim; it does not decide copyright ownership or recover a lost original. The report therefore does not claim a production-ready decentralised rights system.

The financial review materially changes the earlier investment story. Applying the actual plan commissions and processing full buyer payments reduces Year 3 platform revenue to KZT 71.39 million and operating FCF to KZT 5.90 million. Three-year NPV at 25% is KZT -21.43 million. Payback does not occur within three years. The base cash requirement is KZT 35.76 million, leaving a limited buffer in a KZT 40 million funding plan.

Both required pessimistic scenarios create larger financing needs than the proposed raise. The combined scenario is particularly severe. A conditional tranche protects an investor only if management also stops or changes spending when milestones fail. It does not remove the cost of an unsuccessful pilot. These findings connect the software, risk matrix and financial statements to the time value of money and the risk–return trade-off.

The appropriate next step is an evidence-led pilot and a revised investment decision. Validate buyer and creator retention, inspect the contribution after payment fees and support, obtain independent security and legal reviews, and model any reduced scope or repriced funding terms. The current prototype is a suitable teaching demonstration, while the commercial case remains conditional.

The team should support a hypothetical investment only after those conditions are met. For this course, no company needs to be registered, no real funds used and no token issued. Each member must understand the code, calculations and limitations, disclose the AI assistance, document their actual contribution and complete the submission-access checks. A well-supported negative NPV is a valid analytical result and should not be replaced by an invented success claim.

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AI Disclosure Statement

We used Codex with GPT-6.1 Sol for AI assistance in this project. Every additional tool and exact model actually used must also be disclosed before submission.

AI assistance included application and smart-contract code, test code, interface explanations, demonstration tooling, English captions, report editing, reference metadata checks, financial-model code and recalculation, figures and presentation preparation. The revised model applies the team draft's demand and cost anchors, adds an explicit plan mix and corrects payment-fee treatment. These are substantial forms of assistance and must not be described as grammar-only editing.

This disclosure does not claim that students independently produced or already verified every AI-assisted output. Before submission, each member must review the design, assumptions, calculations and conclusions, reproduce the relevant commands, correct any remaining errors and explain any part of the code. Individual self-evaluations must describe actual work and learning in the student's own words.

The financial results remain educational planning estimates. DOI metadata were checked against Crossref. Publisher indexing information supports the journal selection, while the team should retain its own current Scopus source-list checks. No AI chatbot is counted as a literature source. No measured coverage percentage, vendor audit quotation, customer survey or external commercial deployment is invented.

The supplied report and this revision are different evidence sources. This revision follows the current repository and saved local recording rather than the earlier draft's Hardhat commands or hypothetical contract interface. The final submission checklist records the public GitHub URL and the unresolved OneDrive access, individual contributions and student review requirements.

Appendix A Monthly forecast Year 1

Table A1. Monthly base forecast Year 1 in KZT thousand

M	Sales	Revenue	Costs	FCF	Cash
1	0	0	2,411	-2,411	22,589
2	0	0	1,740	-1,740	20,849
3	0	0	1,740	-1,740	19,109
4	0	0	6,353	-6,353	12,756
5	50	156	2,036	-1,880	10,876
6	100	343	2,059	-1,715	9,160
7	150	531	2,081	-1,550	7,610
8	200	719	2,104	-1,385	6,224
9	250	906	2,127	-1,220	5,004
10	300	1,094	2,149	-1,055	18,949
11	350	1,282	2,172	-890	18,058
12	400	1,469	2,195	-725	17,333

Source: Own calculations from scripts/finance-model.mjs. Amounts are hypothetical.

Forecast averages can be fractional before rounding. No opening assets, debt or capitalised development are assumed. All pre-launch expenses enter operating FCF. Cash includes the scheduled equity tranches, while revenue does not. The full unrounded 36-month output is financial-model.json in this package.

Reproduction: node scripts/finance-model.mjs artifacts/submission-review/financial-model.json. Validation: npm run test:finance-model. Five tests check tier weighting, the gross-payment processing base, cash reconciliation, downside behaviour and the two-tranche IRR. These numerical tests verify implementation consistency; they do not validate customer forecasts.

Appendix A Monthly forecast Year 2

Table A2. Monthly base forecast Year 2 in KZT thousand

M	Sales	Revenue	Costs	FCF	Cash
13	475	1,684	3,810	-2,125	15,208
14	550	1,899	3,394	-1,495	13,713
15	625	2,114	3,426	-1,312	12,400
16	700	2,329	4,802	-2,474	9,926
17	775	2,543	3,491	-947	8,979
18	850	2,758	3,523	-765	8,214
19	925	2,973	3,555	-582	7,632
20	1000	3,188	3,588	-400	7,232
21	1075	3,403	3,620	-217	7,015
22	1150	3,617	3,652	-35	6,980
23	1225	3,832	3,685	148	7,128
24	1300	4,047	3,717	330	7,458

Source: Own calculations from scripts/finance-model.mjs. Amounts are hypothetical.

Forecast averages can be fractional before rounding. No opening assets, debt or capitalised development are assumed. All pre-launch expenses enter operating FCF. Cash includes the scheduled equity tranches, while revenue does not. The full unrounded 36-month output is financial-model.json in this package.

Reproduction: node scripts/finance-model.mjs artifacts/submission-review/financial-model.json. Validation: npm run test:finance-model. Five tests check tier weighting, the gross-payment processing base, cash reconciliation, downside behaviour and the two-tranche IRR. These numerical tests verify implementation consistency; they do not validate customer forecasts.

Appendix A Monthly forecast Year 3

Table A3. Monthly base forecast Year 3 in KZT thousand

M	Sales	Revenue	Costs	FCF	Cash
25	1400	4,340	5,519	-1,179	6,279
26	1500	4,632	5,114	-482	5,797
27	1600	4,925	5,157	-232	5,565
28	1700	5,218	6,544	-1,327	4,238
29	1800	5,510	5,244	267	4,504
30	1900	5,803	5,287	516	5,020
31	2000	6,096	5,330	765	5,786
32	2100	6,388	5,374	1,015	6,801
33	2200	6,681	5,417	1,264	8,065
34	2300	6,974	5,460	1,514	9,579
35	2400	7,266	5,503	1,763	11,342
36	2500	7,559	5,546	2,013	13,355

Source: Own calculations from scripts/finance-model.mjs. Amounts are hypothetical.

Forecast averages can be fractional before rounding. No opening assets, debt or capitalised development are assumed. All pre-launch expenses enter operating FCF. Cash includes the scheduled equity tranches, while revenue does not. The full unrounded 36-month output is financial-model.json in this package.

Reproduction: node scripts/finance-model.mjs artifacts/submission-review/financial-model.json. Validation: npm run test:finance-model. Five tests check tier weighting, the gross-payment processing base, cash reconciliation, downside behaviour and the two-tranche IRR. These numerical tests verify implementation consistency; they do not validate customer forecasts.

Appendix B Formulas and valuation

Project cash flows

Commission = sales × average licence price × 6.5%. Subscription revenue = Premium creators × USD 10 + Business creators × USD 50, converted at the frozen planning rate. Processing = 2.5% × (gross licence payments + subscriptions). Operating FCF = revenue − cash costs − planning tax. Financing is separate.

NPV and terminal value

NPV = $FCF_1/1.25 + FCF_2/1.25^2 + FCF_3/1.25^3$. No additional initial investment is subtracted because build expenses already appear in Year 1. Optional TV = max(0, Year 3 FCF) × 1.05 / (0.25 − 0.05). Terminal-value NPV adds $TV/1.25^3$.

Investor calculation

The proposed 26% interest implies post-money value 40/0.26 = KZT 153.85 million. The illustrative exit value is 2.5 × Month 36 revenue × 12 + closing cash = KZT 240.12 million. The multiple is an assumption. Investor proceeds are KZT 62.43 million. IRR solves $-25 - 15/(1+r)^{0.75} + \text{proceeds}/(1+r)^3 = 0$, with all figures in KZT million.

This calculation gives the same final 26% after both tranches and assumes no later dilution, debt, dividends or transaction costs. It is not a promise of return. Downside cases need additional financing, so their investor returns cannot be inferred without modelling the new capital terms.

Tax and timing limits

The simplified model carries operating losses forward and applies the planning 20% rate only after those losses are used. It is not a tax filing or advice on eligibility. Annual NPV uses year-end aggregation; the investor calculation uses explicit tranche dates. Different timing conventions must not be mixed without explanation.

Unit contribution and volume break even

At the assumed tier mix, a licence contributes USD 30 × (6.5% − 2.5%) − USD 0.10 = USD 1.10, or KZT 492.73, before fixed costs. Premium and Business monthly contributions after processing are KZT 4,367.42 and KZT 21,837.08. Required licence sales = max(0, fixed costs − subscription contributions) / 492.73. Fixed costs exclude gas and processing. Apply the subscriber base and cost level of the same month.

Appendix C Actual contract interface

Table C1. Core functions in the supplied repository

Contract or function	Access and effect
ArtworkRegistry.registerArtwork(fileHash, termsHash)	Caller registers a new nonzero fingerprint and terms hash.
EscrowLicenseMarketplace.buyLicenses(quotes, signatures)	Buyer deposits the exact total; validates quotes and opens pending orders.
markDelivered(id)	Seller confirms delivery within 72 hours.
dispute(id)	Buyer disputes during the review window.
release(id)	Buyer accepts, or anyone releases after review expiry if undisputed.
refund(id)	Conditional refund credit under contract state and deadline rules.
resolveDispute(id, refundBuyer)	Immutable arbiter resolves an open dispute.
withdraw(), withdrawRefund()	Caller withdraws its own proceeds or refund credit.
hasLicense(), licenseExpiry(), getEscrow()	Read wallet-bound rights, period and order state.

Source: contracts/ArtworkRegistry.sol and contracts/EscrowLicenseMarketplace.sol.

Table C1 lists the core interface. Build compiler: solc 0.8.37 with optimiser 200 runs, viaIR and Cancun EVM target. Contract pragmas permit compatible 0.8.x versions. OpenZeppelin Contracts 5.6.1 supplies EIP712, ECDSA and ReentrancyGuard under MIT. The separate registry is a supporting contract. The escrow contract is the core purchase use case. Legacy, subscription and promotion contracts do not change that boundary.

Licence duration is 30, 90 or 365 days, or perpetual. An active licence or pending order blocks a repeat purchase of the same licence type for the same wallet and file. A timed licence starts at release and can be purchased again after expiry. It is a stored wallet-bound record, not an NFT or a transferable token.

An EIP-712 quote binds the buyer, seller, file and terms hashes, amount, fee, recipients, duration, expiry and chain instance. The contract checks the recovered signer and registered seller. It trusts the signer for the offered terms rather than judging their legal validity.

Appendix C Test and transaction evidence

The video uses saved run ccab5e4f-561c-473b-b35c-44019c1d437c. Its two test commands passed 21 selected application unit tests and 10 EVM escrow tests, with zero failures and exit code zero. Table C2 summarises the EVM cases. A later client-wide run passed 98 tests. Coverage was not measured and no percentage is supplied.

Table C2. Actual EVM escrow tests

Test	Verified result
1. Deposit holds funds, rejects invalid quotes and checks permissions.	Passed
2. Acceptance allocates exact fees and shares, starts validity and supports withdrawal.	Passed
3. No delivery after 72 hours permits refund credit, withdrawal and repurchase.	Passed
4. Review expiry enables a transaction; API reads do not release funds.	Passed
5. A dispute blocks ordinary release and enforces manager authority.	Passed
6. Manager release and seven-day unresolved-dispute refund follow contract rules.	Passed
7. Bulk deposits settle independently and restart preserves state.	Passed
8. Legacy licence expiry and earlier credits remain readable.	Passed
9. Reconciliation distinguishes locked funds, proceeds credits and refund credits.	Passed
10. Restricted prepared-wallet signer supports the complete escrow scenario.	Passed

Source: tests/escrow.test.ts; saved contracts.log and the subsequent review run.

Direct proof reads the transaction, successful receipt, canonical block, deployed bytecode, decoded events and live contract storage. It distinguishes deposit, release and withdrawal. This proves state changes in the local operator-controlled EVM, not payment on Ethereum mainnet.

Deposit: 0x34ae9d9494b596fbeca51f0a9bd436af51b6e6f6696c054f4898b402b2c3f11d

Release: 0x00cecaaa23803fbdd83f8501d11ac405a4657eb57fce9f52590e634b98f7991e

Withdrawal: 0x580fdebd429abfe00ce413e5c9e16e5d4570f8c412c0f8f711a78e8e2bf1294a

The package includes run.json, on-chain-proof.json and canonical-receipts.json from this run. The MP4 is exactly 180 seconds with English burned-in captions. Seven labelled reading pauses explain recorded evidence without creating new transaction or test results.

Appendix D Full risk register

Table D1. Risk owner, trigger and response

ID	Owner	Trigger and planned response
R1	Assan Dairabaev	Sales below 70% of target for two months; review acquisition and stop the next tranche until justified.
R2	Assan Dairabaev	Customers prefer existing licensing tools; test narrower value proposition.
R3	Demezhan Kaidar	High-severity finding or inconsistent events; stop new quotes and assess replacement deployment.
R4	Demezhan Kaidar	Signer compromise; incident review and a migration design, not fictional key rotation.
R5	Demezhan Kaidar	Gas or outage exceeds budget; defer public-network plan and update the forecast.
R6	Kalybek Zhumabekov	Key member unavailable or backup restore fails; documented handover and recovery exercise.
R7	Kalybek Zhumabekov	Disputes exceed 2% of orders; review evidence policy and support capacity.
R8	Yevgeniy Tyo	Cash below six months of revised costs; reduce commitments and rerun funding scenarios.
R9	Yevgeniy Tyo	USD costs exceed planning FX budget; revise pricing and tools.
R10	Kalybek Zhumabekov	Regulatory classification remains unclear; no real-asset launch before review.
R11	Kalybek Zhumabekov	Data incident or unsuitable transfer; contain access and assess notification obligations.
R12	Kalybek Zhumabekov	Repeated copyright complaints; takedown and appeal review, preserving necessary evidence.

Source: Own planning assessment. Triggers are proposed management thresholds, not observed incidents.

Likelihood and impact are qualitative estimates. Periodic review should update both after pilot research, restore exercises and independent security analysis. Planned controls must not be described as implemented features or guaranteed protections.

Appendix D Risk likelihood and impact map

Figure 4 places all twelve risks on the same 1–5 likelihood and impact scale. The position gives both ratings, and the risk score equals their product. This map supplements the response register in Table D1 and the main risks in Table 4.

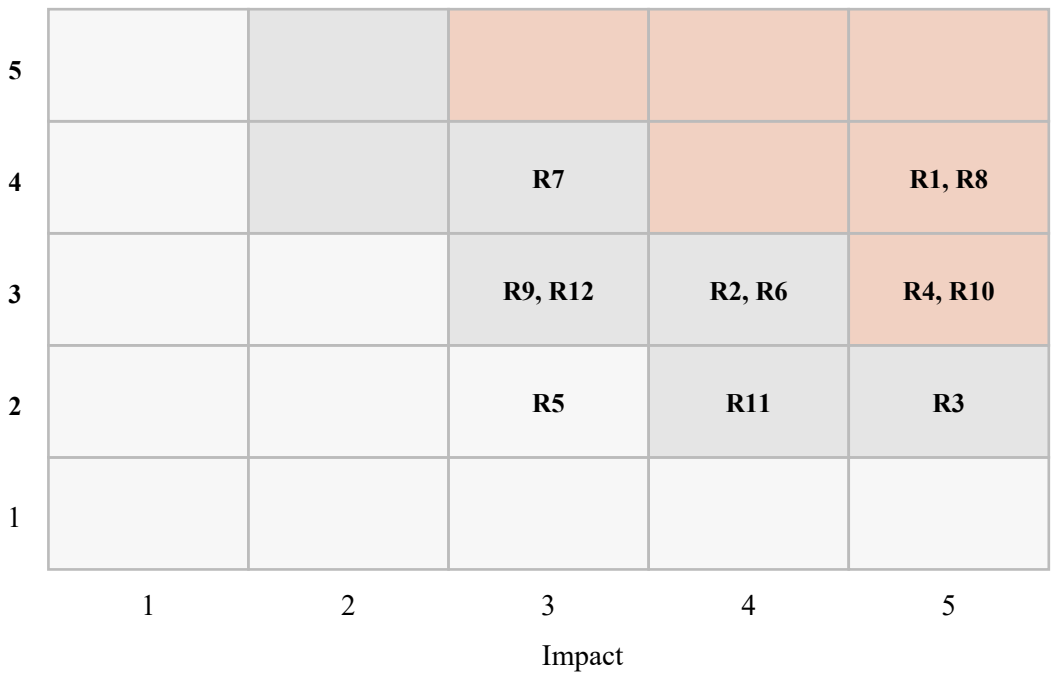


Figure 4. Likelihood and impact map for all twelve risks

Source: Own qualitative assessment. Darker cells indicate higher likelihood × impact scores.

The full ratings are R1 4/5, R2 3/4, R3 2/5, R4 3/5, R5 2/3, R6 3/4, R7 4/3, R8 4/5, R9 3/3, R10 3/5, R11 2/4 and R12 3/3. The order in each pair is likelihood/impact. These values are scenario-planning judgements and should be updated after new evidence.

The high-priority financial and market cells correspond to the downside funding results, not just to software defects. A successful demonstration cannot reduce a demand rating without customer evidence. Similarly, a budgeted hardware wallet does not reduce key risk until a reviewed custody and recovery design is implemented.

Appendix E Literature and source verification

Table E1 maps six 2021–2024 journal articles to this report. DOI titles, authors, years and publication details were checked against Crossref on 8 November 2026. Three papers are in IEEE Access, one in Sensors, one in Future Internet and one in Electronic Markets. At least the first five form the required post-2020 academic set.

Table E1. Academic sources and their specific use

Paper	Use in this report
Lashkari and Musilek (2021)	Consensus and validator trust comparison.
Kushwaha et al. (2022a)	Threat classes and residual contract risk.
Kushwaha et al. (2022b)	Why analysis tools and tests do not replace independent review.
Bhujel and Rahulamathavan (2022)	Difference between an NFT market and a licensing service.
Frattolillo (2024)	Smart contracts and watermarking as complementary rights-management techniques.
Ciriello et al. (2023)	Governance and design issues in digital rights systems.

Source: Verified DOI metadata and the cited publications.

IEEE lists Scopus among its indexing providers (IEEE, n.d.). MDPI lists Sensors and Future Internet among Scopus-covered journals (MDPI, n.d.). These publisher records support selection; the team should retain a current Scopus source-list check by journal and ISSN before submission. This report does not claim that the reviewer logged into an institutional Scopus account.

Frattolillo (2024) studies smart-contract-supported watermarking. ArtGuard uses protected previews and exact-byte hashes, and does not claim to implement the paper's complete watermarking protocol. This distinction matters because a paper about a protection technique is not evidence that this prototype provides the same security guarantees.

Market growth, customer counts, audit budgets, processing rates and the exit multiple remain declared assumptions. No anonymous blog, invented survey, AI chatbot or unsupported market statistic is used as a factual substitute. Retrieved dates describe this review, not evidence of a future production certification.

Appendix F Team and reproducibility

All four members are Software Engineering students in group SE-2533. The project representative confirmed the names and roles in Table 13. The descriptions below identify responsibilities rather than inventing employment experience or qualifications.

Assan Dairabaev. CEO. Review responsibility: Introduction, pitch and conclusion. The member must document actual pages, slides, code changes and learning in their own self-evaluation.

Demezhan Kaidar. CTO. Review responsibility: Architecture, contracts, security and technical evidence. The member must document actual pages, slides, code changes and learning in their own self-evaluation.

Yevgeniy Tyo. CFO. Review responsibility: Financial assumptions, model and sensitivity analysis. The member must document actual pages, slides, code changes and learning in their own self-evaluation.

Kalybek Zhumabekov. COO. Review responsibility: Operations, risk and regulatory analysis. The member must document actual pages, slides, code changes and learning in their own self-evaluation.

The source package contains the application, Solidity contracts, tests, build scripts and this report builder. It excludes accounts, uploads, SQLite databases, runtime keys, .env files and local chain state. OpenZeppelin attribution and licence notices are retained. Each member must contribute through their own GitHub account; no historical authorship or commit history is fabricated.

The public source repository is https://github.com/ONI-COLLI/Artguard. The university OneDrive sharing URL remains pending and must appear on the report title, presentation title and README. The video must use the university account, Can view permissions and the specified audience. Access should be checked by a teammate from their own institutional account before submission.

The team lead submits the report and presentation as PDFs within the 5 MB and 10 MB limits. Each student submits a separate PDF self-evaluation of at most one page and 1 MB. The supplied self-evaluation forms require personal completion; they are not declarations already made by the students.